RandomParadox 0.11.0 Documentation

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July 2026

1 Introduction

A tool to randomly generate scenarios for multiple games.
Currently implemented modules:

- Hearts of iron IV (Scenario and MapTool (Generation, Loadingg))
- Europa Universalis IV !!CURRENT MODS CRASH ON STARTUP!! (MapTool (Generation, Loading))
- Victoria 3 !!CURRENT MODS **MAY** CRASH ON STARTUP OR DURING GENERATION. At **max** you can get into the mapeditor. (Partial Maptool, experimental)

1.1 General Functionality

- Fully random maps with fully random continents, climate, provinces, regions, countries and more
- Generating random terrain and climate from simple input continents
- Generating from detailed input terrain, such as pre-drawn hills, valleys, mountainous areas etc
- Generating from detailed input climate, such as pre-drawn deserts, jungles, boreal areas etc
- Provinces/Region generation, either fully random, or respecting given borders
- Generation of mod files for multiple games, creating loadable mods
- So much more...

1.2 Some general reminders

- You can find help regarding this tool in the discord channel: https://discord.gg/m85TndzufD
- You need to drag in images of the correct resolution
- Keep in mind that you can almost always first have something generated **and find this in the Maps folder next to the executable**, then modify it with your favourite editor, and then drag it in again. This can make your life a lot easier if you want to prepare certain inputs, such as climate, provinces, states or countries (and more)
- It can happen that the tool crashes
- If the tool no longer works like expected, you can click the **reload config** button in the first tab to reset it to the initial configuration
- **For Hoi4**: If you want to play the generated random mods and have things happening, you NEED to install better generic focus tree mods. I also recommend a better ai mod!

To avoid error reports that are due to incorrect configuration, please read the documentation carefully.

2 GUI

2.1 Overview

The GUI consists of a log window, generic settings and buttons, and tabs. The start window will look like this:

At the top, you can see the log messages, keep an eye on them in case of errors. They will only update after calculations are done, so check the console window that opens with the GUI for live updates.



Figure 1: Overview tab

2.2 Configure

In the configure tab, you can set multiple values, and have some automatically set, if possible. First of all, you can select the game you want to generate for, and select a config preset from the list.
There is only one config presets:

- **default**: This preset is meant for the mixture of generating and loading maps, you basically go through all the tabs one by one and generate and load, and configure all the desired settings in one tab

The module should find the mod folder and game folder on startup. If it doesn't, you need to set them and validate the paths.
You can try to find the game via the **Try to find game** button, and try to find the mod via the **Try to find mod folder** button. You can also set the mod name.
Before you can proceed, you will have to successfully validate paths by clicking **Validate all paths**. If it is successful, the other tabs will become clickable.

3 FastWorldGen

3.1 Land Input

This tab gives multiple options to input data for heightmap generation:

- Shape: Just input landshapes, and the tool will automatically generate a heightmap from this. See 3.1.1.
- Heightmap: Just input a more detailed, but still simple heightmap. This gives you a chance to already predefine in a smooth way where mountains will be. The tool will automatically generate a heightmap from this. See **??**.
- Shape: Just input landshapes, and the tool will automatically generate a heightmap from this. See **??**.

3.1.1 Heightmap Generation from Land Mask

Instead of generating a land mass from a heightmap, you may also derive a heightmap from a given land mask. For this, drag in an image. The tool will then automatically try to derive what land is and what is an ocean. If in doubt, try to stick to the colours which are otherwise generated, dark blue or black is considered water.

3.1.2 Heightmap Generation from Landform Input

By selecting the **Landform** mode in the Land Input tab, you enable this feature. Now, when dragging in an image, you can afterwards analyse an image and label every single colour to a land type.
Make sure you have a good image without washed up borders, e.g. don't draw an image with a brush, but use a pencil. You might otherwise get far too many colours to label, this would be tedious.
In this tab, you ONLY label the input, then proceed to the heightmap tab for the generation. Always click **Analyse** after labeling terrain, only then is it applied for the heightmap generation!

3.2 Heightmap

Here, you can generate a heightmap or load one by dragging in an image. You can also skip this step if you have a terrain map that you want to input, you can do this in the **Land Input** tab.
When generating an image, you have a large amount of settings available and can also manipulate settings for individual layers of the heightmap.

- **SeaLevel**: The target sealevel. The heightmap will be adjusted to match this, please note that for the different games these values are hardcoded, only change if you know what you are doing.
- **LandPercentage**: The target share of land
- **Height Adjustments**: Negative numbers lower the heightmap, positive numbers increase it
- **Landlayer** coastal distance factor: When land layers are applied to the generated heightmap, they keep a distance to the coast, from which on they decrease, to avoid mountains directly at the coasts. The higher, the larger the distance
- **Maximum lake size factor**: Determines which seabodies will be classified as lakes, e.g. the larger this is, the larger bodies of water can be before detected as lakes.
- **Maximum land height**: Determines the maximum height of the heightmap on land. Must be larger than sealevel. If equal to sealevel + 1, the map will be completely flat.

3.2.1 Random heightmap generation

Generating a random heightmap is a two step process. First of all, you generate the basic shape of the land using **Generate land shape from**

seed. When you're happy with that shape, you can next click on **Generate detailed heightmap from shape**. This will generate more detail on top of the shape.
You may also do this in a one step click using **Generate complete random heightmap from new seed**.
If you tick **Change seed on every generation click**, it will automatically change every time you either generate a land shape or the land detail.

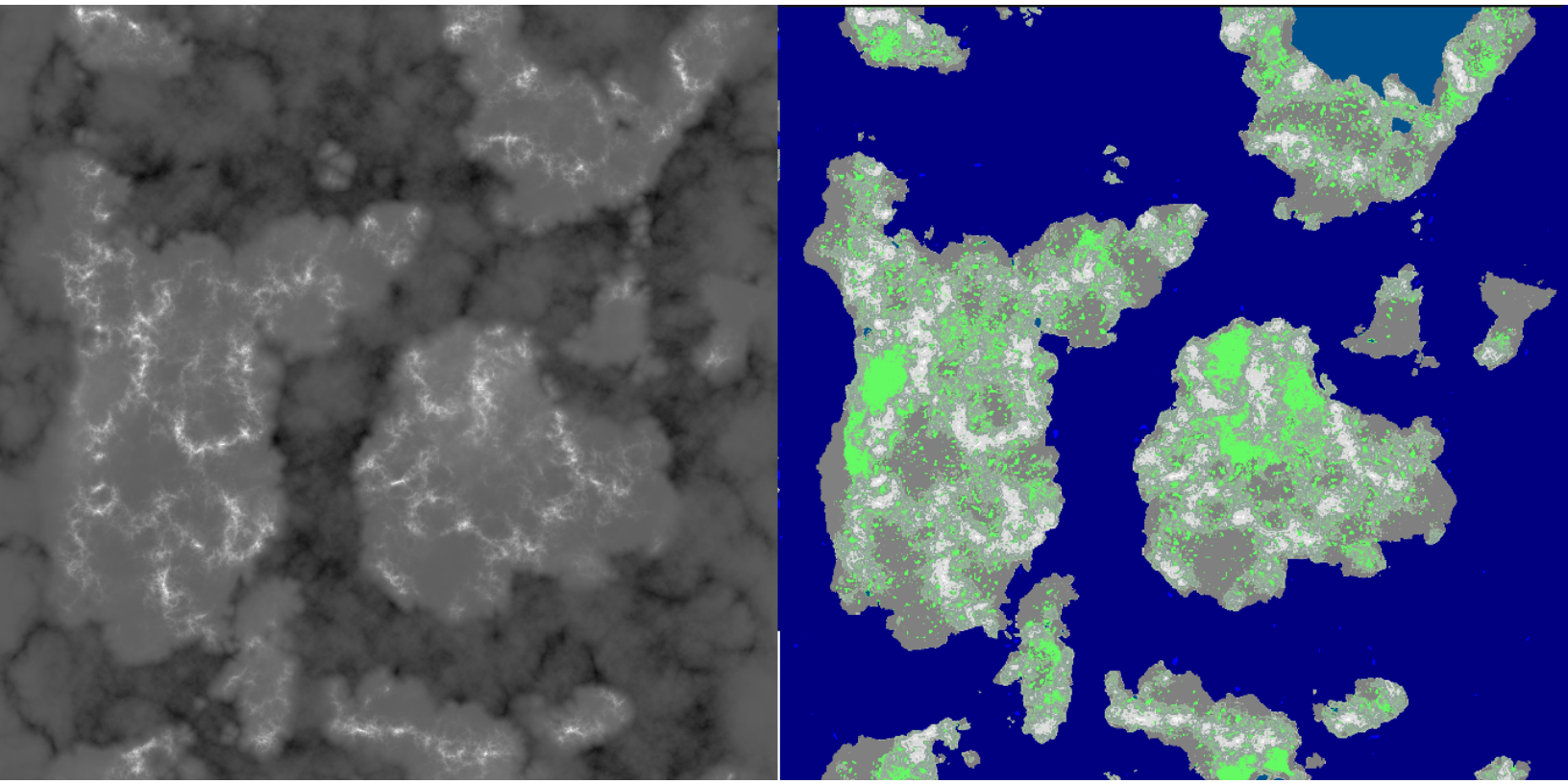


Figure 2: A random heightmap and the corresponding land shape classification.

3.2.2 Individual layer settings

Heightmaps are generated from multiple combined layers. To edit these layers, open the "Layer Editor". **Please note that this is pretty advanced and not necessary to edit in most cases!** There are three different types of layers, Shape (for basic continent shape), Land (to give continents details), and sea (to give oceans more details).

Each layer has different properties:

- **type:** the fractal type that is used
- **landLayer:** if this layer will be applied only to land after the basic continent shape has already been determined
- **fractalFrequency:** the higher, the smaller the fractals
- **fractalOctaves:** affects ruggedness
- **fractalGain:** how rugged the generation is
- **seed:** individual seed for a layer, if 0, it is automatically determined based on the base seed
- **Edge Fade Width:** from when on terrain height falls towards the east/west edges of the map
- **Edge Fade Height:** from when on terrain height falls towards the north/south edges of the map
- **weight:** how much this single layer is weighted relative to the other layers
- **maxHeight:** what the total maximum height present in this layer can be. Should be between 1 and 255
- **minHeight:** what the total maximum height present in this layer can be. Should be between 1 and 255, and smaller maxHeight
- **tanFactor:** how quickly land height increases. Can produce really rugged layers, specifically good for e.g. mountain ranges. Range between 0.00 and 0.26

3.2.3 Heightmap generation from land shape input

Here is the explanation of how input works. Your input defines the overall distinction between land and water, and then the different types affect how much elevation will be randomly generated. This is done via land layers and sea layers. You can edit these layers if you extend the layer editing. Furthermore, each type has a modifier for average altitude, overlaid over a noisemap that is built from the land layers in the heightmap layer config. This means, that the higher the input factor, the more likely you will actually see this terrain at this location. These modifiers are relative to each other and can be modified. Any elevation generation will be normalised to the range 0-maximumLandHeight.
See figure 3 for an example.

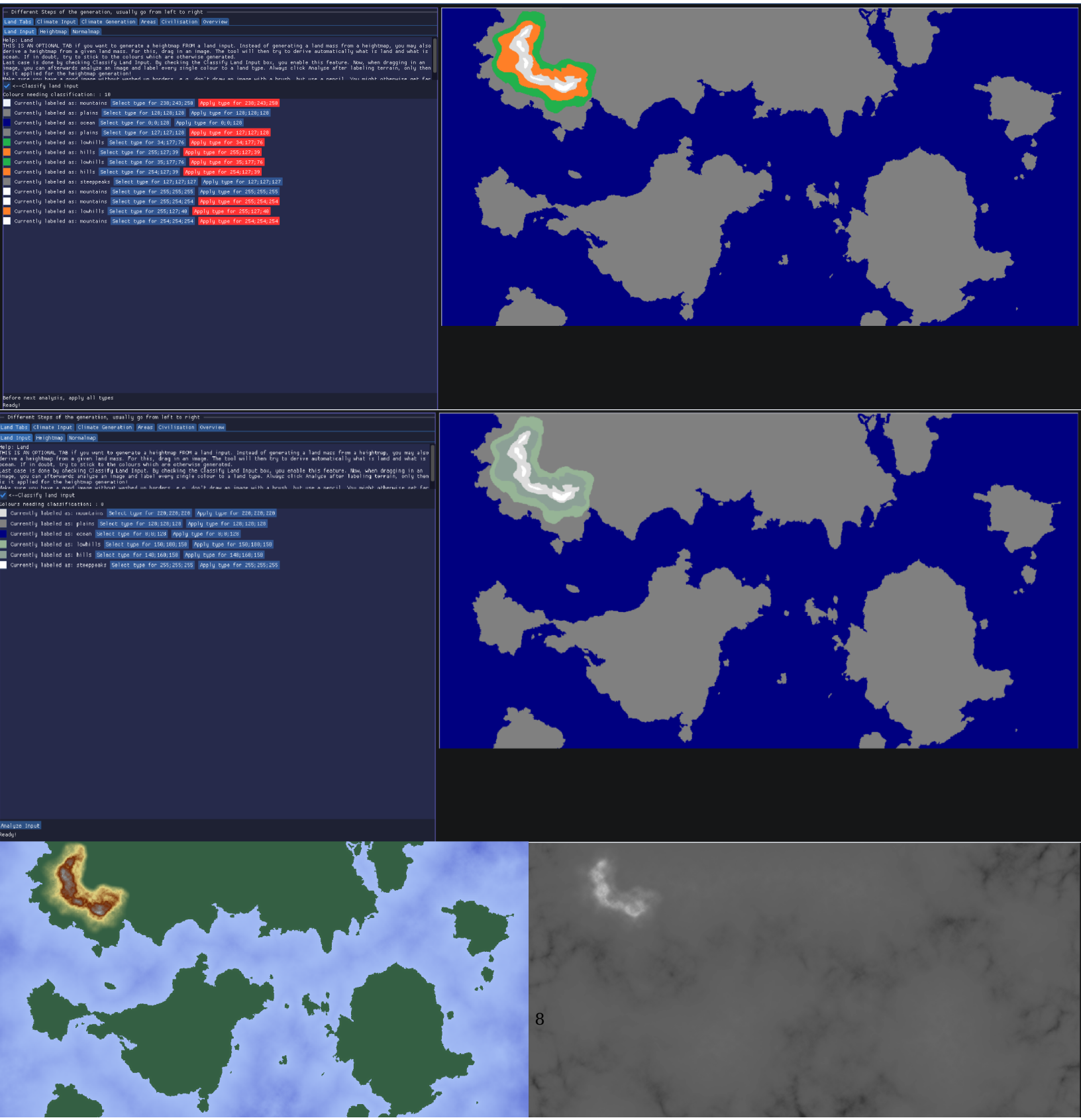


Figure 3: The dragged in input while labelling the colours, and the output

3.3 Normalmap

You can generate a normal map in this tab. Modifying the **SobelFactor** makes the land bumpier or smoother. For an example see figure 4.



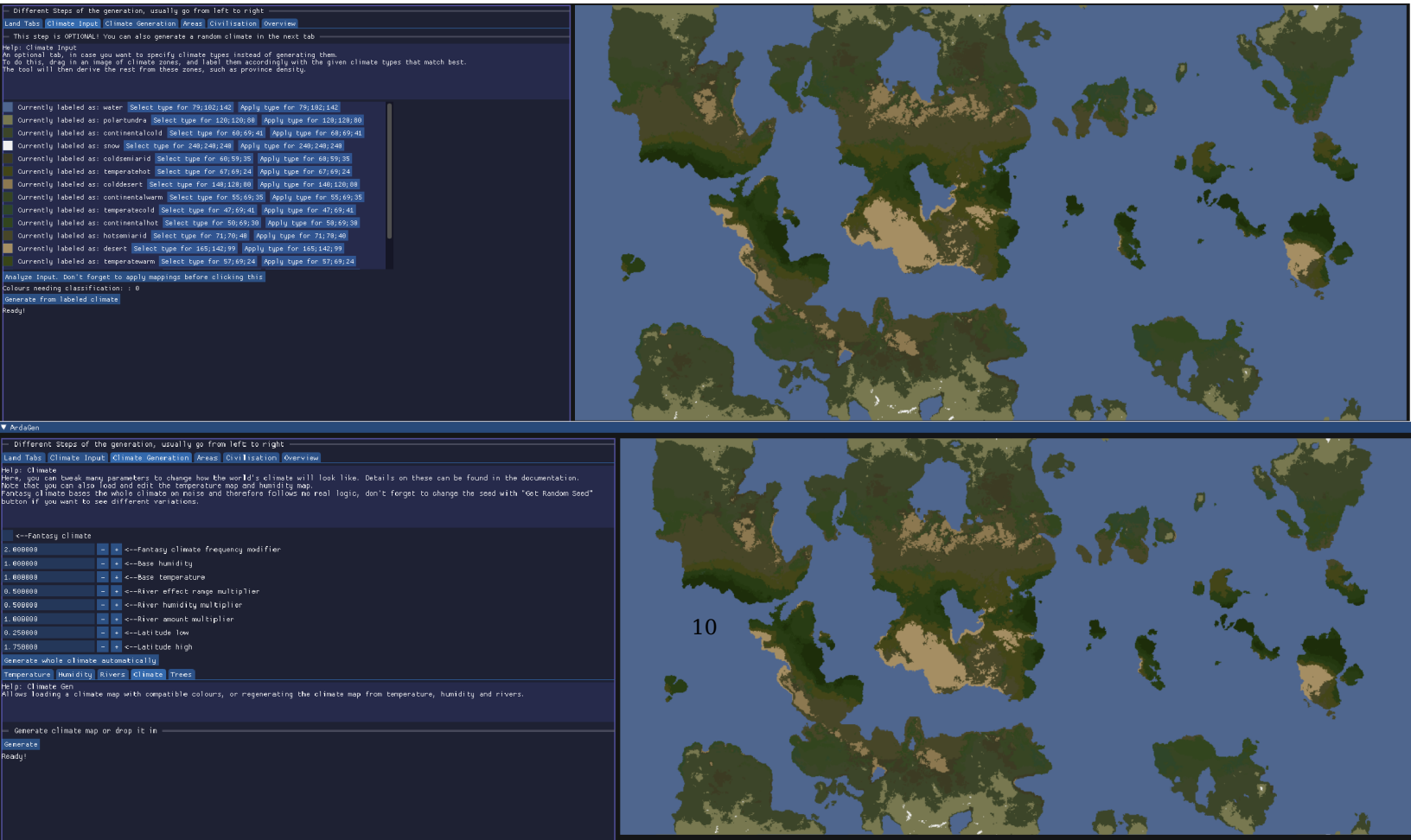
Figure 4: Generated Normal Map

3.4 Climate Input

An optional tab, in case you want to specify climate types instead of generating them. To do this, drag in an image of climate zones, and label them accordingly with the given climate types that match best. The tool will then derive the rest from these zones, such as province density, see 5.
Important note: There is also another way to input data for climate generation, by inputting data into the temperature and humidity subtabs in the Climate tab, see 3.5.1 and 3.5.2.
This is the specification of climates:

Name	Humidity	Temp	Altitude	Habitability	Tree Factor	Colour (RGB)	Tree Type	Arability
tropicsrainforest	1.0	0.93	0.0	0.2	0.36	29-45-10	TROPICALMOIST	0.5
tropicsmonsoon	0.9	0.9	0.0	0.6	0.54	35-55-15	TROPICALMOIST	0.6
tropicssavanna	0.7	0.8	0.0	0.7	0.32	40-60-20	TROPICALDRY	0.7
desert	0.1	0.9	0.0	0.0	0.0	165-142-99	NONE	0.0
colddesert	0.1	0.5	0.0	0.05	0.04	140-120-80	NONE	0.0
hotsemiarid	0.2	0.7	0.0	0.3	0.24	71-70-40	SPARSE	0.6
coldsemiarid	0.2	0.4	0.0	0.3	0.16	60-59-35	SPARSE	0.6
temperatehot	0.45	0.7	0.0	0.7	0.24	67-69-24	TEMPERATEMIXED	0.9
temperatewarm	0.45	0.6	0.0	0.7	0.36	57-69-24	TEMPERATEMIXED	1.0
temperatecold	0.5	0.5	0.0	0.6	0.56	47-69-41	TEMPERATENEEDLE	0.8
continentalhot	0.4	0.6	0.0	0.5	0.4	50-69-30	TEMPERATEMIXED	0.9
continentalwarm	0.4	0.5	0.0	0.6	0.49	55-69-35	TEMPERATENEEDLE	1.0
continentalcold	0.4	0.4	0.0	0.5	0.72	60-69-41	BOREAL	0.7
polartundra	0.4	0.1	0.0	0.05	0.0	120-120-80	NONE	0.1
polararctic	0.4	0.05	0.0	0.0	0.0	240-240-240	NONE	0.0
snow	0.0	0.1	0.2	0.0	0.0	240-240-240	NONE	0.0
water	0.1	0.0	0.0	0.0	0.0	79-102-142	NONE	0.0

Table 1: Biome data including humidity, temperature, altitude, habitability, tree factor, RGB colour, tree type, and arability. They define how climates are chosen from the different values, and what they define if loaded and labeled.



3.5 Climate Generation

This Tab has multiple sub tabs. However, you can automatically generate everything by clicking **Generate Climate Automatically**. The factors to modify are:

- **Base humidity**: How humid the overall world is. The lower, the drier.
- **Base temperature**: How hot the overall world is. The lower, the more ice. The higher, the more jungle/desert
- **Fantasy Climate Frequency Modifier**: The higher this is, the smaller each random area is
- **Fantasy Climate**: This creates completely random climate, which follows no real world rules
- **River effect range multiplier**: How wide the effect of rivers on humidity and therefore climate is
- **River humidity multiplier**: How strong the effect of rivers on humidity and therefore climate is
- **River amount multiplier**: Affects how many rivers there are
- **Latitude high**: The upper bound of the latitude. 2.0=90 degrees north (North pole). Equator at 1.0
- **Latitude low**: The lower bound of the latitude. 0.0=90 degrees south (south pole). Equator at 1.0

For an example, see figure 6.

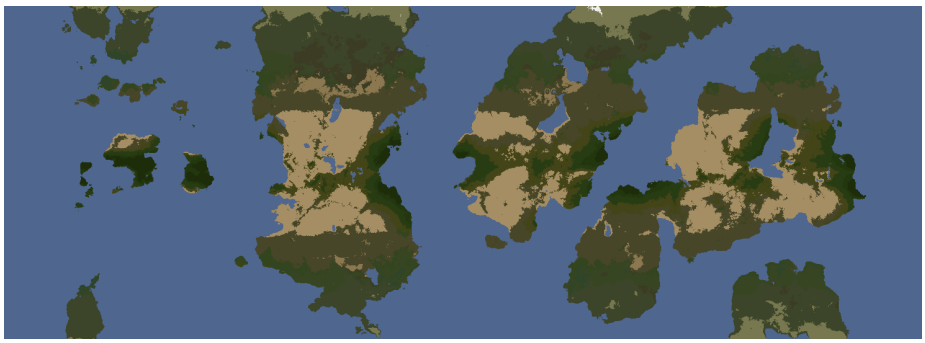


Figure 6: Generated Climate Map

3.5.1 Temperature tab

Temperature is a major factor for climate generation. If you don't want to handle climate zones in detail, you can simply drop in a temperature map as greyscale and generate climate classification based on it.
If you check **Apply elevation effect when loading**, the elevation will have an influence on the temperature. This allows to input a more simple map while still getting the detailed output that elevation produces.

3.5.2 Humidity tab

Humidity is a major factor for climate generation. If you don't want to handle climate zones in detail, you can simply drop in a humidity map as greyscale and generate climate classification based on it.
If you check **Apply elevation effect when loading**, the elevation will have an influence on the humidity. This allows to input a more simple map while still getting the detailed output that elevation produces.

3.5.3 Rivers

You can modify a factor on how many rivers you want. Rivers also affect climate generation, see the factors listed above in the climate overview. Rivers can be loaded, here is the list of colour inputs and their meaning:

- River Colour: 0, 0, 255
- River Start: 0, 255, 0
- Tributary Start: 0, 128, 0
- River End: 255, 100, 0

For Hearts of Iron IV, you may also load a river map with the default colours from the vanilla rivers.bmp. However, **every single river needs a river start pixel**! E.g. loading the vanilla Hoi4 map will only load parts of the rivers, if no river start pixels are added!

3.5.4 Climate

In the default "Simulated Climate" mode, you generate climate according to the parameters at the top.
You can switch to the Random Climate mode, to generate climate based only on noise.
To input Climate, use the Climate Input Mode

3.5.5 Forests

Here you can create a forest map. This will be overlaid over the climate zone map. You can increase density of trees in different zones with the density factors listed here.

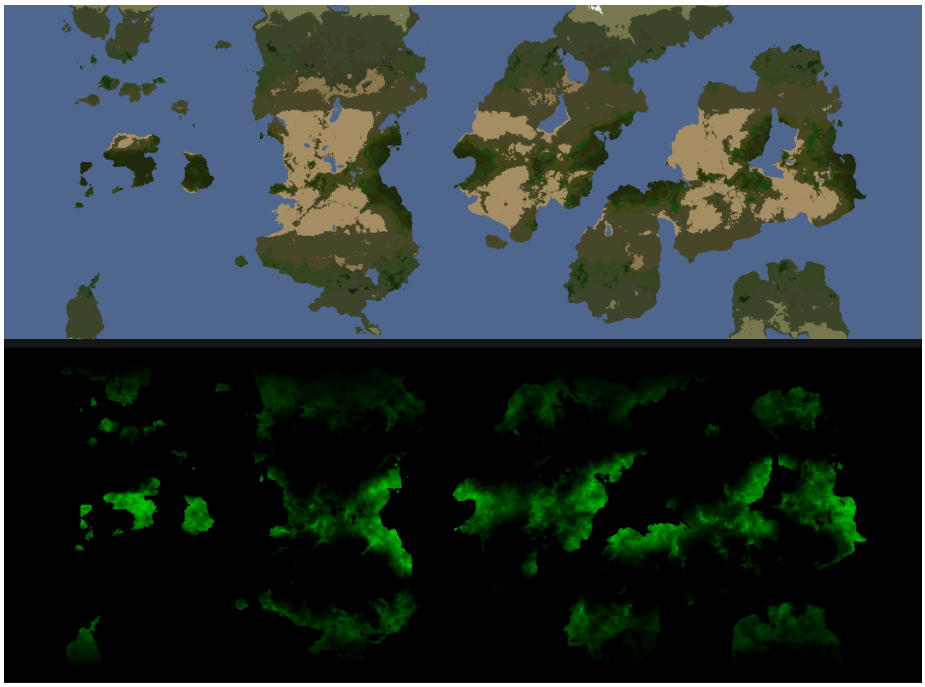


Figure 7: Visualisation of Tree Coverage on climate map

3.6 Areas

3.6.1 Area Input Modes

The tabs SuperSegments, Segments and Provinces allow two different types of input.

- Solid colour input with or without black (rgb 0,0,0) borders: These black borders are ignored and filled, they do not constitute an area. Having borders first helps creating a solid colour input image.
- Borders: This can be activated by clicking on the Borders input type radio button. Here, this loads anything that is between borders and coasts as an area. Using the **Generate supersegment/segment/province template to draw in** button, you can get three different templates in

the Maps/areas folder of the tool. When you draw in these, you can create simple black (rgb 0,0,0) borders, which will be supersegment/segment/provinces borders.
This whole process allows you to craft very detailed country/state/province borders in a multi step process, as the tempaltes area based on each other. E.g. when you have finished drawing the supersegment borders and have loaded them, you can then generate a segment template that already has these borders in them, to craft your segment/state borders in there, load it, and proceed to provinces.

You may stop this process after every step, and use the default generation!
As an example, please see 8

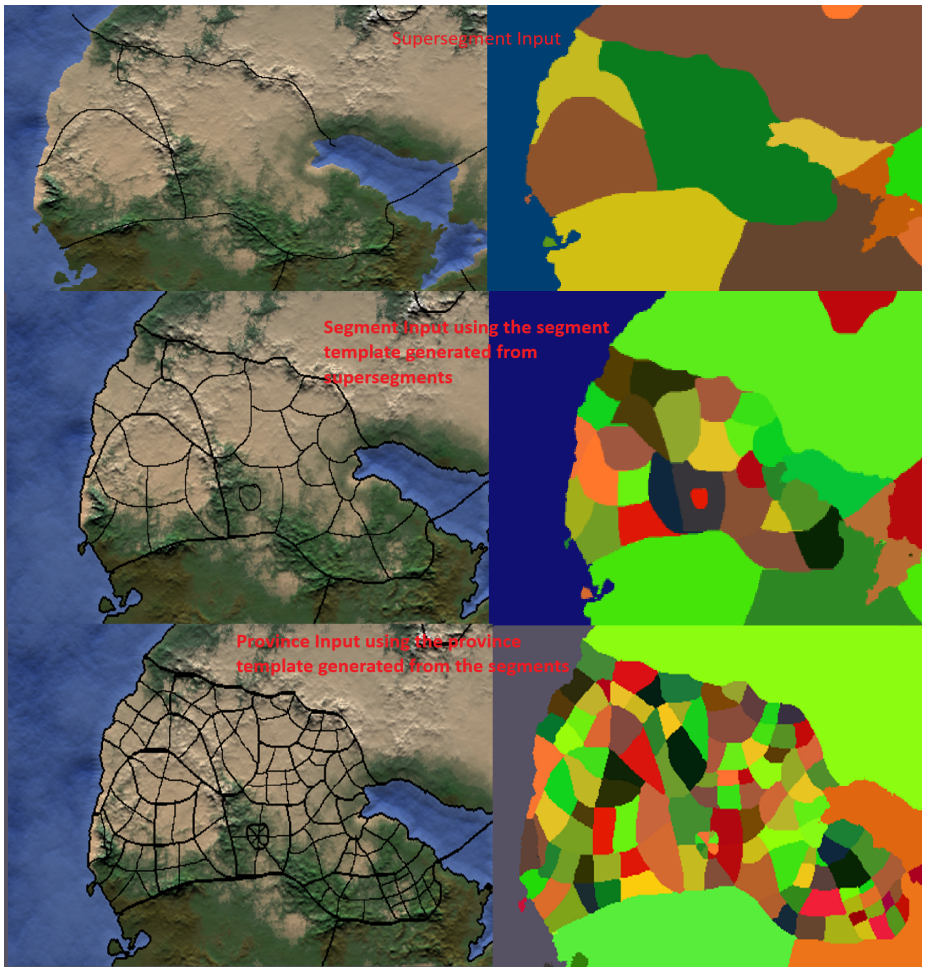


Figure 8: The process to craft detailed areas. You may stop this process after every step, and use the default generation.

3.6.2 Density

Either automatically calculate the density of provinces and population from the climate, or drag and drop in a custom density map. White means dense, small provinces, black means large provinces. You may also simply use the auto generated one, which is based upon the habitability of the climate region.

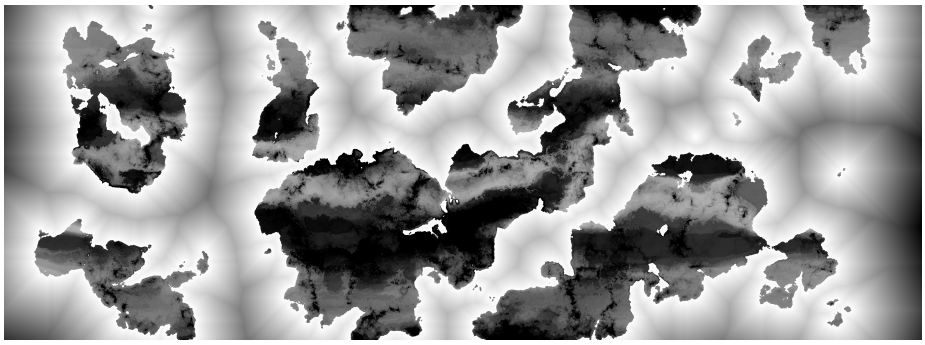


Figure 9: Generated Density Map

3.6.3 SuperSegments

Supersegments define the areas in which segments are generated. They are by default detected from land- and waterbodies, but can be loaded. Loaded supersegments are especially useful if you want to predefine larger areas for segments, such as countries. When loading supersegments, please note the different input types, see 3.6.1.
As Supersegments influence segment generation, and segment generation influences province generation, Supersegments define clear borders for provinces.

3.6.4 Segments

Segments define the areas in which provinces are generated. This tab allows generating and loading them. When loading segments, please note the different input types, see 3.6.1.
To generate, click the **Generate Segments** button, see 10. You may change settings:

- **Segment Cost Influence**: How much the terrain and climate influences the shape of the segments.
- **targetLandRegionAmount**: How many segments there should be on land
- **targetSeaRegionAmount**: How many segments there should be on ocean/lakes

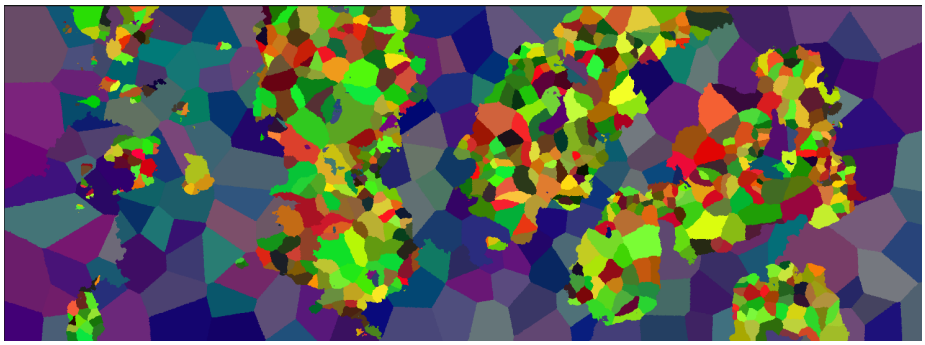


Figure 10: A visualisation of generated segments overlaid over the worldmap.

To load, you have to drag in an image, which defines areas by their colour, so you may for example drag in a state map (looks exactly like a generated segment map), and provinces will be generated only inside the borders of this state map, depending on the density.

3.6.5 Provinces

Here you can configure province generation and generate provinces until you are happy with the result, or load a province map. When loading provinces, please note the different input types, see 3.6.1. Most important factors are:

- **Landprovincefactor**: The higher, the more provinces on land
- **Seaprovincefactor**: The higher, the more provinces on oceans
- **Density Effects**: The higher, the more the density map influences density and size of provinces
- **Minimum size of provinces**: The higher, the more pixels a province will have
- **Maximum amount of provinces**: An upper limit to province amount, increasing this drastically can have major effects on the generated game modules. **For Hearts of Iron IV, this upper limit should not be larger than 20000, otherwise visual bugs start to appear!**

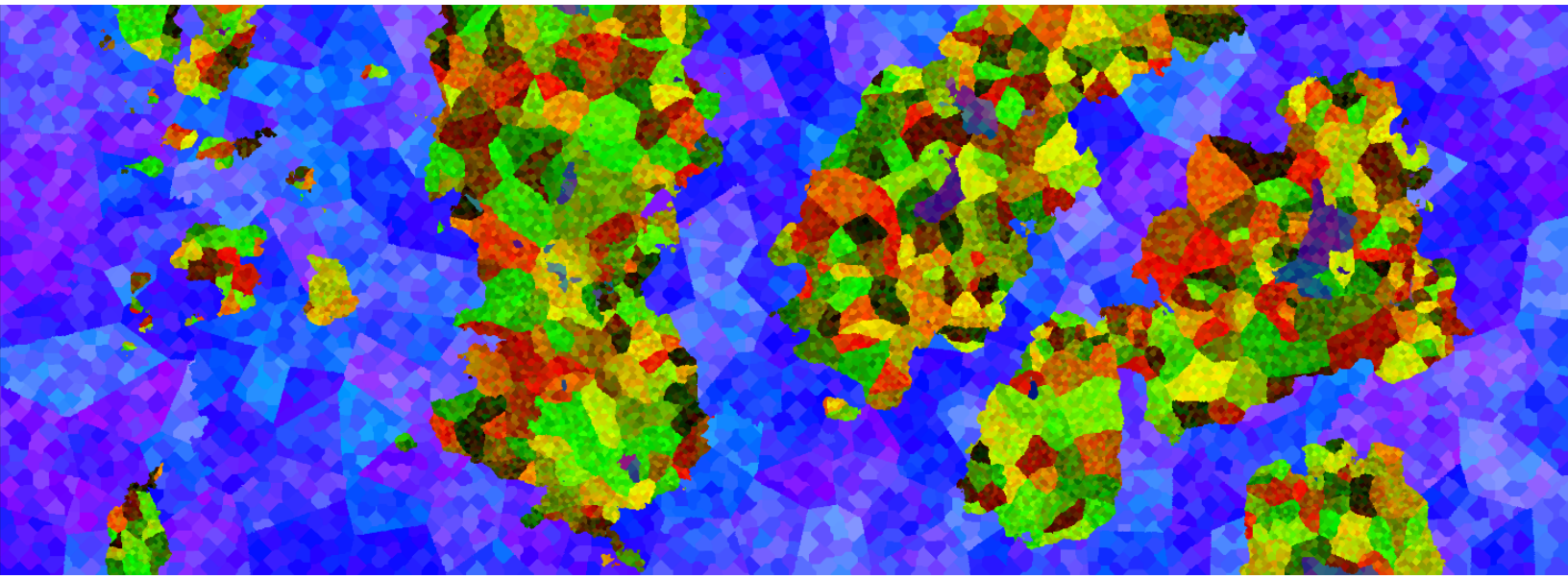


Figure 11: Generated Provinces Map

3.6.6 Continents

This shows an overview of the continents that were detected. You can also input a map of continents yourself. Any colour on water is ignored, but do NOT use black (rgb 0,0,0) as a colour, only as borders! They are detected from larger landbodies, with smaller bodies being assigned to their nearest continent. You can define the desired amount of continents. For an example see figure 12.



Figure 12: Generated Continent Map

3.7 Civilisation

Here you can generate and edit the development, population and location distribution in three sub-tabs:

- **Topography**: Allows dropping in zones that are marshes or wastelands. Doesn't do much else yet.
- **Development**: Affects industrialisation and cities.
- **Population**: Affects population amount and city size. You can draw values for single provinces here, using the brush settings at the top.
- **Culture**: Affects languages and names of countries, states, locations, characters and any later generation that uses language data.
- **Locations**: This displays where cities, farms, mines and more are generated on the world.

3.7.1 Topography

Dragging in an image, it allows to paint in marsh and wasteland.

- Marsh: Used by the modules. Input colour is rgb 40, 255, 40
- Wasteland: Used by the modules. Input colour is rgb 255, 0, 0

3.7.2 Development

You may load a development map as greyscale input. There is two ways to generate development:

- Generate Development with new random base development modifiers: Randomly sets a modifier per continent, then applies habitability and other factors on top to calculate the development of a province
- Generate Development with current base development modifiers: Randomly sets a modifier per continent, then applies habitability and other factors on top to calculate the development of a province. **You may modify the development modifier** of a continent, by clicking it on the map and settings it modifier at the bottom of the tab.

When turned off, you can manually set the Continent Development modifiers. Development is still modified by climate, so increasing it to 1.0 does not mean completely developed continents, but the factor can be higher than 1.0.

3.7.3 Population

Generate population automatically and draw in this map. You may also load a population map.

3.7.4 Culture

You can currently not load cultures.
Cultures define the languages and the people of an area. They are used to generate names for countries, states, characters etc. You may currently only change the nameset used for random name generation by clicking on a culture colour in the image, and then on edit language.
Culture loading will follow.

3.7.5 Locations

You can modify some parameters:

- **Amount of separate cities per region**: The higher, the more distinct cities there are.
- **Amount of separate farm areas per region**: The higher, the more distinct farms there are

3.8 Scenario

This tab allows setting of options for the selected game module. Depending on the game, different options show up. This tab contains the Remap areas button, which is important if you go back to the previous tabs and change something there. Make sure to remap then before proceeding to countries!

3.9 Countries

This tab allows configuration of the number of countries you want generated. You can also select 0. You can also input a list of countries and a country image.

3.9.1 Country Text Input

The list of countries has the format **r;g;b;tag;name;adjective**. If a country has no input mapping, a random country will be generated for it. You can also export the current state to the Maps//areas folder, for safekeeping. This is also a good basis for editing this list, as it gives you the already existing colours and tags already in a list in the right format.

3.9.2 Country Image Input

This tab also allows inputting in two different forms, similar to the areas.
Recommendation: It is often beneficial to use a two step process: First use the border input, drag in a border image. This will have islands etc as their own countries.
However, the exported countries.png in the Maps/areas folder will contain a solid colour country image. Edit this using the "bucket fill tool" of common image editors, then switch to the solid mode and load this new image, having distributed land bodies however you like.
You can input a country image, the states will be assigned as good as possible to match the borders of the countries. Make sure that you have unique colours for countries, so when drawing an image, **don't use a brush, use a pencil!**. You can also first generate a country map and find it in the **Maps** folder, then edit it with your favourite image editor, and then drag it in again. You can also use country input in the barriers tab.

3.9.3 Country editing

All the loaded or generated countries can be edited. See 13 for the buttons that pop up.

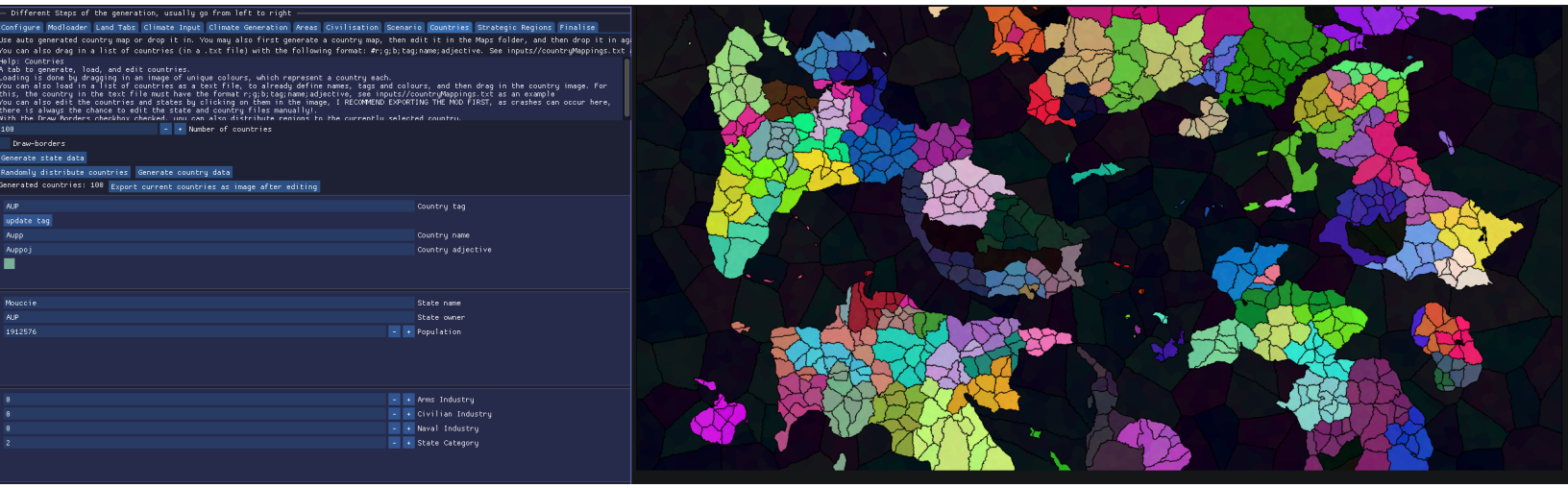


Figure 13: Generated Countries Map

3.10 Strategic regions

You can generate, load or edit strategic regions with the drawing functionality in this tab. This tab also allows inputting in two different forms, similar to the areas. There is one key difference in the Borders input: White borders are in the template, they are ignored in the image. Coasts do not count as borders!!! You need to instead draw lines there as well. This is because strategic regions often contain small water bodies or islands.